

VINCE GONZALEZ

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Formal Methods | Proof Assistant Tooling | Software

I contribute to formal mathematics libraries and build the instruments that measure them. Nine of my changes are merged into set.mm. Every tool I ship reports what it cannot model instead of guessing.

FORMAL MATHEMATICS

set.mm, Metamath main library | Contributor | 2026

- Nine merged contributions (PRs 5442, 5443, 5445, 5446, 5447, 5448, 5458, 5466, 5469).
- Several remove an avoidable axiom dependency from an existing proof, which requires rebuilding the proof without the axiom rather than restating the theorem.
- One merge freed 53 downstream results from a choice dependency.

mmforge | Author | PyPI, MIT

- Finds avoidable axiom dependencies in Metamath databases and builds the proofs that remove them.
- Its statement parser missed 24 statements whose typecode sits on the following line. Correcting it moved the measured graph from 50,662 labels and 1,507,069 edges to 50,686 and 1,508,483, and moved the reach of the choice axioms from 523 to 524.

mmrewrap | Author | MIT

- Pure-Python reimplement of Metamath comment rewrapping, removing a hard dependency on the compiled C binary for contributors with no local build.
- Measured against the whole library: 50,554 of 50,685 statement comments byte-exact, 100.0000% of the population it accepts, with a mechanical refusal gate naming all 4,801 comments it does not model.

rocqaxioms | Author | MIT

- Source-only axiom readers for Rocq, Agda, Isabelle and the AFP. No build, no toolchain.
- Separates mathematical assumptions from interface obligations. A Parameter inside a Module Type is discharged on instantiation; a census that counts it reports a library assuming hundreds of things it does not assume.

metamath-exe | Open pull request

- PR 203, under maintainer review. metamath has segfaulted on 64-bit Windows since 2023 (issue 164, reported by Mario Carneiro). I diagnosed the specific mechanism, that pointer alignment pads the null pnterString block so its length field is read out of position, and wrote the fix. With it, 33 of 41 tests pass on 64-bit, against zero before.

PUBLISHED

Cross-foundation axiom census | Zenodo, CC-BY-4.0 | [concept DOI 10.5281/zenodo.22148932](https://doi.org/10.5281/zenodo.22148932)

- Rocq stdlib, Corelib, MathComp and the Agda standard library measured with one instrument, every corpus commit pinned so the numbers reproduce.

SOFTWARE

F-Keys Creative LLC | Founder | 2021-Present

- 47 packages published across PyPI and npm in Python and JavaScript.
- OpticQuiz, a color vision deficiency engine implemented from the primary literature, implemented twice in two languages with a paired test suite holding both to identical output to six decimal places.
- Builds and runs opticquiz.com, trailer-load.com, poticas.com and 5best2buy.com end to end. Release gates fail the build.

EXPERIENCE

Federal Express Corporation | Operations Supervisor | 2020-Present

- Supervise a sort operation of up to 30 people against a fixed dispatch time.
- Promoted to Operations Supervisor in May 2023. New hire facilitator for the outbound sort.

Coastal Pacific Wine and Spirits Brokers | Combo Sales Rep | 2015-2019

- Managed 100+ on- and off-premise accounts. Ranked 15th of more than 400 representatives across 17 control states.

EDUCATION

Ohio University | B.A. Philosophy, Pre-Law, minor in History | 2012

Formal logic sequence. Gateway Scholarship Award.

Columbus State Community College | Computer Science coursework | 2010-2011; 2022-2023

TECHNICAL

Metamath, Rocq, Agda, Isabelle/AFP. Python, JavaScript, Node.js, Postgres, Cloudflare Workers and D1, REST APIs, Git/GitHub, GitHub Actions, CI gates, automated testing, Linux, local LLM deployment.